Literature Status Note: Question 7.3 of arXiv:1908.07814

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2026-07-19

Status. literature_already_answered. This is a literature-status packet, not an original proof packet.

Source question. The source paper is Li–Nowak–Spakula–Zhang, *Quasi-local Algebras and Asymptotic Expanders*, arXiv:1908.07814 [1]. In Section 7, Question 7.3 asks whether, for a coarse disjoint union X of bounded-geometry asymptotic expanders $\{X_n\}_{n\in\mathbb{N}}$, the space X can coarsely embed into a Hilbert space. The source paper proves a negative answer only with the extra hypothesis $C_u^*(X) = C_{uq}^*(X)$ in Proposition 7.4.

Answering paper. The supporting paper is Khukhro–Li–Vigolo–Zhang, *On the structure of asymptotic expanders*, arXiv:1910.13320 [2]. The introduction explicitly identifies [21, Question 7.3] as the motivating question and says that the paper gives a negative answer. Theorem B, restated as Theorem 4.2, proves the stronger statement that if $\{X_n\}_{n\in\mathbb{N}}$ is a bounded-geometry sequence of finite metric spaces with $|X_n| \to \infty$, X is its coarse disjoint union, and $\{X_n\}$ is a sequence of asymptotic expanders, then X cannot be coarsely embedded into any Banach space in the class $\mathcal{C}_0$. This class includes Hilbert-space targets, and the paper records the corresponding obstruction for L^p spaces, $1 \le p < \infty$, and uniformly curved Banach spaces.

Identification. The source question asks for the existence of a Hilbert coarse embedding under bounded geometry and the asymptotic-expander hypothesis. The answer paper uses the same objects and explicitly cites the source question. Theorem 4.2 negates the existence of such embeddings into a larger class of Banach targets, so it answers Question 7.3 negatively.

Scope limitations. This packet records only the already-known negative answer to Question 7.3 of [1]. It does not claim a new proof. It also does not settle Question 7.1 from the source paper. The supporting paper has additional applications to the coarse Baum–Connes conjecture, but those are outside this packet's scoped claim.

Search evidence. The run's cheap indexes had no exact existing packet for arXiv:1908.07814 or Question 7.3. Web/arXiv search located arXiv:1910.13320, whose introduction and Theorem 4.2 explicitly identify and answer [21, Question 7.3]. This packet therefore belongs in the literature-status folder, not in `solutions/full/`.

References

[1] Kang Li, Piotr Nowak, Jan Spakula, and Jiawen Zhang, *Quasi-local Algebras and Asymptotic Expanders*, arXiv:1908.07814, 2019. https://arxiv.org/abs/1908.07814

[2] Ana Khukhro, Kang Li, Federico Vigolo, and Jiawen Zhang, *On the structure of asymptotic expanders*, arXiv:1910.13320, 2019; Advances in Mathematics 393 (2021), 108073. https://arxiv.org/abs/1910.13320